

Partial Fractions and Improper Integrals

ANSWER KEY



Partial fraction decomposition: distinct linear factors

- 1 Decompose $\frac{5x-1}{(x-1)(x+2)}$ into partial fractions.
 $\frac{5x-1}{(x-1)(x+2)} = \frac{A}{x-1} + \frac{B}{x+2}$. $5x-1 = A(x+2) + B(x-1)$. At $x=1$: $4 = 3A \Rightarrow A = \frac{4}{3}$. At $x=-2$: $-11 = -3B \Rightarrow B = \frac{11}{3}$. So $\frac{4/3}{x-1} + \frac{11/3}{x+2}$.
- 2 Evaluate $\int \frac{3x+5}{x^2+x-2} dx$ using partial fractions (note $x^2+x-2 = (x+2)(x-1)$).
 $\frac{3x+5}{(x+2)(x-1)} = \frac{A}{x+2} + \frac{B}{x-1}$. $3x+5 = A(x-1) + B(x+2)$. At $x=1$: $8 = 3B \Rightarrow B = \frac{8}{3}$. At $x=-2$: $-1 = -3A \Rightarrow A = \frac{1}{3}$. $\int (\frac{1/3}{x+2} + \frac{8/3}{x-1}) dx = \frac{1}{3} \ln|x+2| + \frac{8}{3} \ln|x-1| + C$.
- 3 Decompose $\frac{x+7}{x^2-x-6}$ into partial fractions (note $x^2-x-6 = (x-3)(x+2)$).
 $\frac{x+7}{(x-3)(x+2)} = \frac{A}{x-3} + \frac{B}{x+2}$. $x+7 = A(x+2) + B(x-3)$. At $x=3$: $10 = 5A \Rightarrow A = 2$. At $x=-2$: $5 = -5B \Rightarrow B = -1$. So $\frac{2}{x-3} - \frac{1}{x+2}$.

Partial fractions with a repeated linear factor

- 4 Decompose $\frac{2x+3}{(x-1)^2}$ into partial fractions.
 $\frac{2x+3}{(x-1)^2} = \frac{A}{x-1} + \frac{B}{(x-1)^2}$. $2x+3 = A(x-1) + B$. At $x=1$: $5 = B$. Comparing x coefficients: $A = 2$. So $\frac{2}{x-1} + \frac{5}{(x-1)^2}$.
- 5 Evaluate $\int \frac{2x+3}{(x-1)^2} dx$ using the decomposition found above.
 $\int (\frac{2}{x-1} + \frac{5}{(x-1)^2}) dx = 2 \ln|x-1| - \frac{5}{x-1} + C$.

Partial fractions with an irreducible quadratic factor

- 6 Set up (but do not solve for constants) the partial fraction decomposition of $\frac{3x^2+2}{(x-1)(x^2+4)}$.
Since x^2+4 is irreducible over the reals, the decomposition has the form $\frac{A}{x-1} + \frac{Bx+C}{x^2+4}$.

Improper integrals: infinite limits

- 7 Evaluate $\int_1^{\infty} \frac{1}{x^2} dx$, or show it diverges.
 $\int_1^{\infty} \frac{1}{x^2} dx = \lim_{b \rightarrow \infty} [-\frac{1}{x}]_1^b = \lim_{b \rightarrow \infty} (-\frac{1}{b} + 1) = 0 + 1 = 1$. Converges to 1.
- 8 Evaluate $\int_1^{\infty} \frac{1}{x} dx$, or show it diverges.
 $\int_1^{\infty} \frac{1}{x} dx = \lim_{b \rightarrow \infty} [\ln x]_1^b = \lim_{b \rightarrow \infty} (\ln b - 0) = \infty$. Diverges.
- 9 Evaluate $\int_0^{\infty} e^{-2x} dx$, or show it diverges.
 $\int_0^{\infty} e^{-2x} dx = \lim_{b \rightarrow \infty} [-\frac{1}{2} e^{-2x}]_0^b = \lim_{b \rightarrow \infty} (-\frac{1}{2} e^{-2b} + \frac{1}{2}) = 0 + \frac{1}{2} = \frac{1}{2}$. Converges to $\frac{1}{2}$.

- 10 Evaluate $\int_{-\infty}^0 e^x dx$, or show it diverges.
 $\int_{-\infty}^0 e^x dx = \lim_{a \rightarrow -\infty} [e^x]_a^0 = \lim_{a \rightarrow -\infty} (1 - e^a) = 1 - 0 = 1$. Converges to 1.

Improper integrals: discontinuous integrands

- 11 Evaluate $\int_0^1 \frac{1}{\sqrt{x}} dx$, or show it diverges (note the integrand is undefined at $x = 0$).
 $\int_0^1 x^{-1/2} dx = \lim_{a \rightarrow 0^+} [2\sqrt{x}]_a^1 = \lim_{a \rightarrow 0^+} (2 - 2\sqrt{a}) = 2 - 0 = 2$. Converges to 2.
- 12 Evaluate $\int_0^1 \frac{1}{x} dx$, or show it diverges.
 $\int_0^1 \frac{1}{x} dx = \lim_{a \rightarrow 0^+} [\ln x]_a^1 = \lim_{a \rightarrow 0^+} (0 - \ln a) = \infty$. Diverges.
- 13 Evaluate $\int_{-2}^2 \frac{1}{x-2} dx$, or show it diverges (the integrand is undefined at $x = 2$).
 $\int_0^2 (x-2)^{-2} dx = \lim_{b \rightarrow 2^-} \left[-\frac{1}{x-2}\right]_0^b = \lim_{b \rightarrow 2^-} \left(-\frac{1}{b-2} - \frac{1}{2}\right) = \infty$. Diverges.

The p-test for improper integrals

- 14 State the p-test for $\int_1^{\infty} \frac{1}{x^p} dx$, and use it to determine convergence for $p = 3$ and $p = 0.5$.
 $\int_1^{\infty} \frac{1}{x^p} dx$ converges if $p > 1$ and diverges if $p \leq 1$. For $p = 3 > 1$: converges. For $p = 0.5 \leq 1$: diverges.
- 15 State the analogous p-test for $\int_0^1 \frac{1}{x^p} dx$ (a discontinuity at $x = 0$), and determine convergence for $p = \frac{1}{2}$ and $p = 2$.
 $\int_0^1 \frac{1}{x^p} dx$ converges if $p < 1$ and diverges if $p \geq 1$ (opposite condition from the infinite-limit case). For $p = \frac{1}{2} < 1$: converges. For $p = 2 \geq 1$: diverges.

Combining partial fractions with improper integrals

- 16 Evaluate $\int_0^{\infty} \frac{1}{(x+1)(x+2)} dx$ using partial fractions.
 $\frac{1}{(x+1)(x+2)} = \frac{1}{x+1} - \frac{1}{x+2}$. $\int_0^b \left(\frac{1}{x+1} - \frac{1}{x+2}\right) dx = \left[\ln \frac{x+1}{x+2}\right]_0^b = \ln \frac{b+1}{b+2} - \ln \frac{1}{2}$. As $b \rightarrow \infty$, $\frac{b+1}{b+2} \rightarrow 1$, so the limit is $\ln 1 - \ln \frac{1}{2} = \ln 2$. Converges to $\ln 2$.

Three distinct linear factors

- 17 Decompose $\frac{6x^2 - 3x - 1}{x(x-1)(x+1)}$ into partial fractions.
 $\frac{6x^2 - 3x - 1}{x(x-1)(x+1)} = \frac{A}{x} + \frac{B}{x-1} + \frac{C}{x+1}$. $6x^2 - 3x - 1 = A(x-1)(x+1) + Bx(x+1) + Cx(x-1)$. At $x = 0$: $-1 = -A \Rightarrow A = 1$. At $x = 1$: $2 = 2B \Rightarrow B = 1$. At $x = -1$: $8 = 2C \Rightarrow C = 4$. So $\frac{1}{x} + \frac{1}{x-1} + \frac{4}{x+1}$.
- 18 Evaluate $\int \frac{6x^2 - 3x - 1}{x(x-1)(x+1)} dx$ using the decomposition above.
 $\int \left(\frac{1}{x} + \frac{1}{x-1} + \frac{4}{x+1}\right) dx = \ln|x| + \ln|x-1| + 4\ln|x+1| + C$.

Visualizing convergence: area under the curve

- 19** The convergent improper integral $\int_1^{\infty} \frac{1}{x^2} dx = 1$ (shown earlier) means the infinite region under $y = \frac{1}{x^2}$ for $x \geq 1$ has FINITE area. Explain intuitively why this is possible even though the region extends forever.
- Even though the region is infinitely long, the height $\frac{1}{x^2}$ shrinks fast enough as $x \rightarrow \infty$ that the accumulated area still converges to a finite total — the 'tail' contributes less and less area, summing to a bounded amount, similar to how a convergent geometric series sums to a finite value despite infinitely many terms.
- 20** By contrast, $\int_1^{\infty} \frac{1}{x} dx$ diverges. Compare the graphs of $y = \frac{1}{x}$ and $y = \frac{1}{x^2}$ on $[1, 10]$ and explain why $\frac{1}{x}$ decays too slowly for the area to stay finite.
- Both curves decrease, but $\frac{1}{x}$ stays larger than $\frac{1}{x^2}$ for all $x > 1$ (e.g. at $x = 10$: 0.1 vs 0.01). This slower decay means the accumulated area under $\frac{1}{x}$ keeps growing without bound as $x \rightarrow \infty$, while $\frac{1}{x^2}$'s faster decay allows its total area to converge.
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Improper integrals with both an infinite bound and a discontinuity

- 21** Evaluate $\int_0^{\infty} \frac{1}{\sqrt{x}(1+x)} dx$ is a classic doubly-improper integral (discontinuous at $x = 0$ AND infinite upper bound). Explain how such an integral must be split to evaluate it.
- Split at any convenient interior point, e.g. $x = 1$: $\int_0^{\infty} = \int_0^1 + \int_1^{\infty}$. Each piece is evaluated as its own limit (one with $a \rightarrow 0^+$, the other with $b \rightarrow \infty$), and the original integral converges only if BOTH pieces converge.
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Comparison test for improper integrals

- 22** Use a direct comparison to argue that $\int_1^{\infty} \frac{1}{x^2+1} dx$ converges, without computing it exactly.
- For $x \geq 1$: $\frac{1}{x^2+1} < \frac{1}{x^2}$. Since $\int_1^{\infty} \frac{1}{x^2} dx$ converges (shown earlier, $p = 2 > 1$), and the given integrand is smaller and positive, $\int_1^{\infty} \frac{1}{x^2+1} dx$ also converges by comparison.
- 23** Use a direct comparison to argue that $\int_1^{\infty} \frac{2+\sin x}{x} dx$ diverges.
- Since $-1 \leq \sin x \leq 1$, we have $2 + \sin x \geq 1$, so $\frac{2+\sin x}{x} \geq \frac{1}{x}$ for $x \geq 1$. Since $\int_1^{\infty} \frac{1}{x} dx$ diverges, and the given integrand is at least as large, $\int_1^{\infty} \frac{2+\sin x}{x} dx$ also diverges by comparison.